

Financial News Sentiment Prediction using Deep Learning & BERT

Two-page model comparison report

1. Dataset and Methodology

The project uses the Twitter Financial News Sentiment dataset with 9,938 training instances and 2,486 validation instances. Tweets are cleaned for URLs, mentions and whitespace while financial tickers are preserved. The baseline vocabulary is built only from training data. Two recurrent baselines, Simple RNN and LSTM, are trained and compared with a fine-tuned ProsusAI/FinBERT model.

2. Quantitative Comparison

Model	Accuracy	Macro F1	Precision	Recall
RNN	78.48%	66.54%	69.50%	64.67%
LSTM	81.24%	72.14%	74.66%	70.36%
FINBERT	87.77%	83.72%	83.42%	84.05%

3. RNN vs LSTM Interpretation

The Simple RNN and LSTM use the same overall embedding-to-classifier pipeline, making their validation comparison meaningful. LSTM adds gated recurrent memory, which is designed to retain useful information over longer sequences and reduce problems associated with basic recurrent state updates. The observed metric difference is reported as an empirical result of this experiment, rather than as proof of a single causal factor.

4. FinBERT Comparison

FinBERT starts from a pretrained transformer representation and is fine-tuned for the three financial sentiment classes. The comparison table reports its measured validation performance against the recurrent baselines using the same official validation split.

5. Class-wise Performance

RNN

Class	Precision	Recall	F1	Support
Bearish	58.23%	41.79%	48.66%	347
Bullish	65.52%	60.00%	62.64%	475
Neutral	84.74%	92.21%	88.32%	1566

LSTM

Class	Precision	Recall	F1	Support
Bearish	62.27%	58.50%	60.33%	347
Bullish	75.53%	59.79%	66.75%	475
Neutral	86.18%	92.78%	89.36%	1566

FINBERT

Class	Precision	Recall	F1	Support
Bearish	75.55%	79.25%	77.36%	347
Bullish	82.48%	81.26%	81.87%	475
Neutral	92.22%	91.63%	91.93%	1566

6. Error Analysis

The training scripts save representative misclassified validation examples. These files contain the original tweet, actual label and predicted label, allowing qualitative discussion of ambiguous financial language, neutral-versus-directional wording, ticker-heavy text, and other recurring failure patterns observed in the validation predictions.

Representative FinBERT errors:

Actual: Bearish **Predicted:** Neutral

Barclays assigns only a 20% chance that studies on a Gilead antiviral drug being done in China will succeed against... <https://t.co/lIgDhy1Gli>

Actual: Bearish **Predicted:** Neutral

BTIG points to breakfast pressure for Dunkin' Brands

Actual: Bullish **Predicted:** Neutral

Peloton bulls defend the upside case

Actual: Bullish **Predicted:** Neutral

Piper Jaffray analyst Erinn Murphy reiterated an Overweight rating and \$44.00 price target on Crocs (NASDAQ: \$CROX)...

<https://t.co/iVlcTMWm1r>

7. Streamlit Application

The Streamlit dashboard supports RNN, LSTM and FinBERT inference. Users can enter a financial tweet or headline and receive the predicted sentiment, confidence, class probabilities and probability bar chart. The benchmark tab also displays validation class distribution, model accuracy and macro-F1 comparisons, and confusion matrices.